

EXP 5: Clustering with K-Means and Dimensionality Reduction with PCA

Aim

To apply **K-Means clustering** for grouping pet adoption data and **Principal Component Analysis (PCA)** for dimensionality reduction and visualization.

Algorithm

K-Means Clustering

1. Import the required libraries.
2. Load the Pet Adoption dataset.
3. Handle missing values.
4. Encode categorical variables.
5. Standardize the features.
6. Apply the Elbow Method to determine the suitable number of clusters.
7. Apply K-Means clustering.
8. Assign each pet to a cluster.
9. Visualize the clusters.
10. Calculate the Silhouette Score.

PCA

1. Standardize the dataset.
2. Apply PCA to reduce the dimensions to two principal components.
3. Calculate the explained variance.
4. Visualize the reduced dataset using PC1 and PC2.
5. Apply K-Means on the PCA-reduced data.
6. Calculate the Silhouette Score.

Code:

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns

from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.cluster import KMeans

from sklearn.decomposition import PCA

from sklearn.metrics import silhouette_score

df = pd.read_csv("pet_adoption.csv")

print("Dataset loaded successfully!")

print("Dataset Shape:", df.shape)

display(df.head())

df["Breed"] = df["Breed"].fillna(df["Breed"].mode()[0])

df["Vaccinated"] = df["Vaccinated"].fillna(df["Vaccinated"].mode()[0])

df["Fee"] = df["Fee"].fillna(df["Fee"].median())

X = df.drop("AdoptionSpeed", axis=1)


categorical_cols = X.select_dtypes(

    include=["str", "object"]

).columns

X_encoded = X.copy()

for col in categorical_cols:

    le = LabelEncoder()

    X_encoded[col] = le.fit_transform(

        X_encoded[col].astype(str)

    )
```

```
scaler = StandardScaler()

X_scaled = scaler.fit_transform(X_encoded)

print("\nData preprocessing completed!")

print("Processed Data Shape:", X_scaled.shape)

wcss = []

for k in range(1, 11):

    kmeans = KMeans(

        n_clusters=k,

        init="k-means++",

        n_init=10,

        random_state=42

    )

    kmeans.fit(X_scaled)

    wcss.append(kmeans.inertia_)

plt.figure(figsize=(8, 5))

plt.plot(

    range(1, 11),

    wcss,

    marker="o"

)

plt.title("Elbow Method for Optimal K")

plt.xlabel("Number of Clusters")
```

```
plt.ylabel("WCSS")

plt.grid(True)

plt.show()

optimal_k = 3

kmeans = KMeans(

    n_clusters=optimal_k,

    init="k-means++",

    n_init=10,

    random_state=42

)

clusters = kmeans.fit_predict(X_scaled)

df["Cluster"] = clusters

print("\nK-Means Clustering completed!")

print("\nCluster Counts:")

print(df["Cluster"].value_counts())

pca_visual = PCA(n_components=2)

X_pca = pca_visual.fit_transform(X_scaled)


pca_df = pd.DataFrame(

    X_pca,

    columns=["Principal_Component_1",

            "Principal_Component_2"]

)
```

```
pca_df["Cluster"] = clusters

plt.figure(figsize=(8, 6))

sns.scatterplot(

    data=pca_df,

    x="Principal_Component_1",

    y="Principal_Component_2",

    hue="Cluster",

    palette="viridis",

    s=70

)

plt.title("K-Means Clusters - PCA Visualization")

plt.xlabel("Principal Component 1")

plt.ylabel("Principal Component 2")

plt.grid(True)

plt.show()

silhouette_avg = silhouette_score(

    X_scaled,

    clusters

)

print("\nSilhouette Score:")

print(round(silhouette_avg, 4))

pca = PCA(n_components=2)

principal_components = pca.fit_transform(
```

```
    X_scaled
)
explained_variance = (
    pca.explained_variance_ratio_
)
print("\nExplained Variance Ratio:")
print(explained_variance)
print(
    "\nTotal Explained Variance:",
    round(explained_variance.sum(), 4)
)
pca_result = pd.DataFrame(
    principal_components,
    columns=[
        "Principal_Component_1",
        "Principal_Component_2"
    ]
)
plt.figure(figsize=(8, 6))
sns.scatterplot(
    data=pca_result,
    x="Principal_Component_1",
    y="Principal_Component_2",
```

```
s=70

)

plt.title(

    "PCA - Dimensionality Reduction"

)

plt.xlabel(

    f"PC1 ({explained_variance[0] * 100:.2f}%)"

)

plt.ylabel(

    f"PC2 ({explained_variance[1] * 100:.2f}%)"

)

plt.grid(True)

plt.show()

kmeans_pca = KMeans(

    n_clusters=optimal_k,

    init="k-means++",

    n_init=10,

    random_state=42

)

clusters_pca = kmeans_pca.fit_predict(

    principal_components

)

pca_result["Cluster"] = clusters_pca
```

```
plt.figure(figsize=(8, 6))

sns.scatterplot(

    data=pca_result,

    x="Principal_Component_1",

    y="Principal_Component_2",

    hue="Cluster",

    palette="viridis",

    s=70

)

plt.title(

    "K-Means Clustering on PCA-Reduced Data"

)

plt.xlabel("Principal Component 1")

plt.ylabel("Principal Component 2")

plt.grid(True)

plt.show()

silhouette_pca = silhouette_score(

    principal_components,

    clusters_pca

)

print(

    "\nSilhouette Score after PCA:",

    round(silhouette_pca, 4)
```

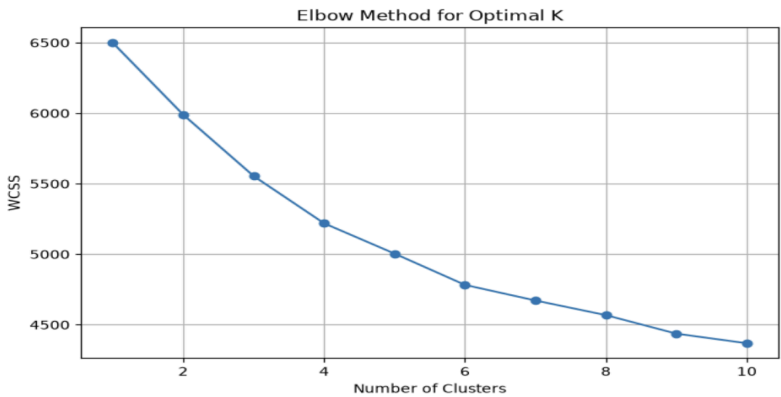
```
)  
  
print("Dataset Size:", df.shape)  
  
print("Number of Clusters:", optimal_k)  
  
print(  
    "K-Means Silhouette Score:",  
    round(silhouette_avg, 4)  
)  
  
print(  
    "PCA Explained Variance:",  
    round(explained_variance.sum(), 4)  
)  
  
print(  
    "PCA K-Means Silhouette Score:",  
    round(silhouette_pca, 4)  
)
```

Output:

Dataset loaded successfully!
Dataset Shape: (500, 14)

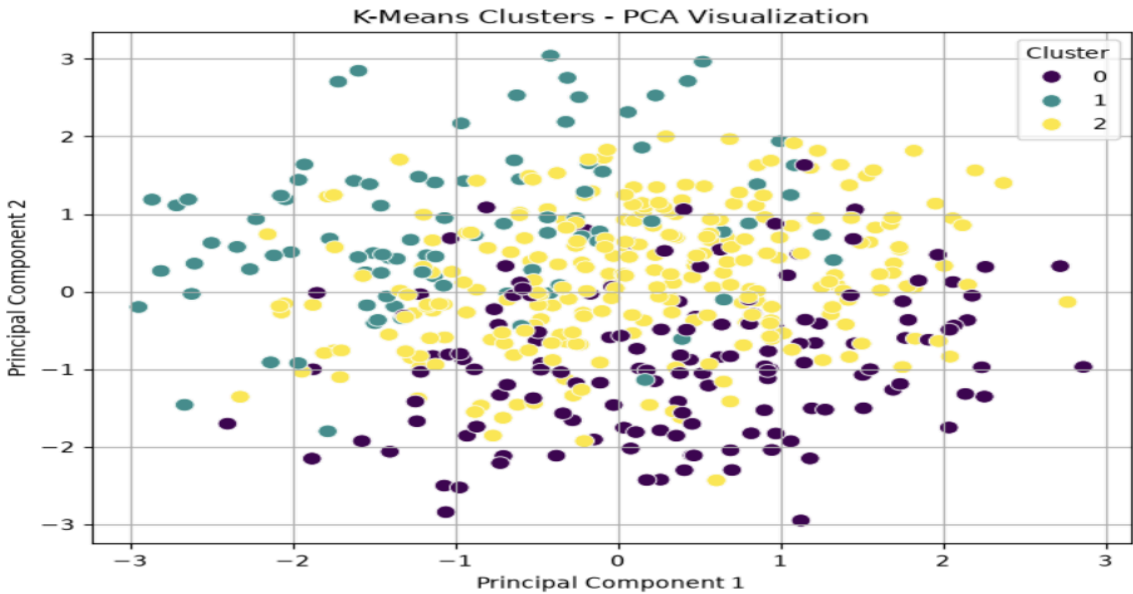
PetType	AgeMonths	Gender	Breed	Vaccinated	Dewormed	Sterilized	Health	MaturitySize	FurLength	Fee	PhotoAmount	VideoAmount	AdoptionSpeed
Dog	64	Male	Beagle	Yes	No	Yes	Healthy	Small	Medium	732.0	4	0	1
Cat	86	Male	Persian	No	Yes	No	Healthy	Large	Medium	1748.0	6	0	2
Dog	33	Female	Mixed Breed	No	Yes	Yes	Healthy	Medium	Medium	3924.0	9	2	3
Dog	88	Male	Beagle	No	Yes	Yes	Healthy	Small	Medium	423.0	10	0	3
Dog	34	Male	Mixed Breed	Yes	Yes	Yes	Healthy	Large	Short	3478.0	10	1	2

Data preprocessing completed!
Processed Data Shape: (500, 13)



K-Means Clustering completed!

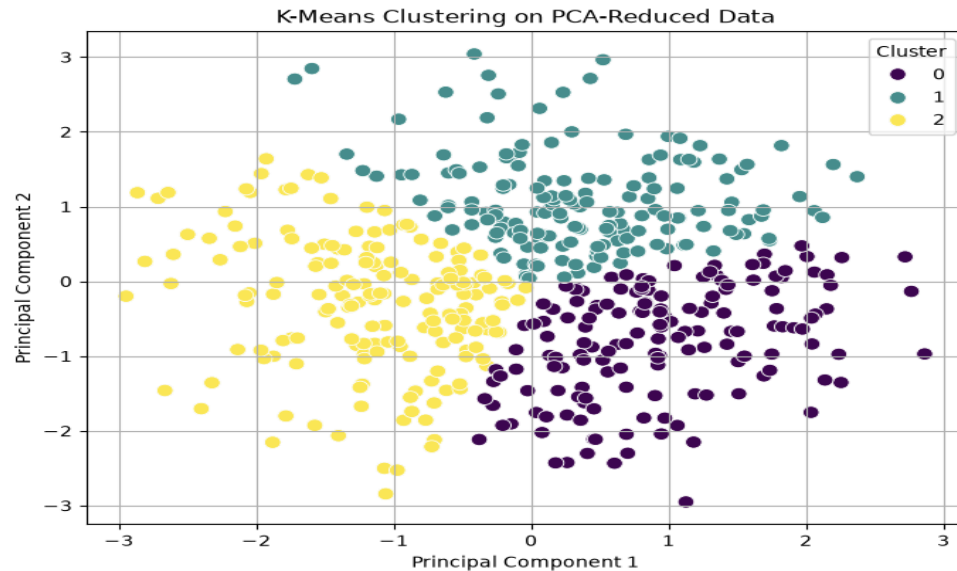
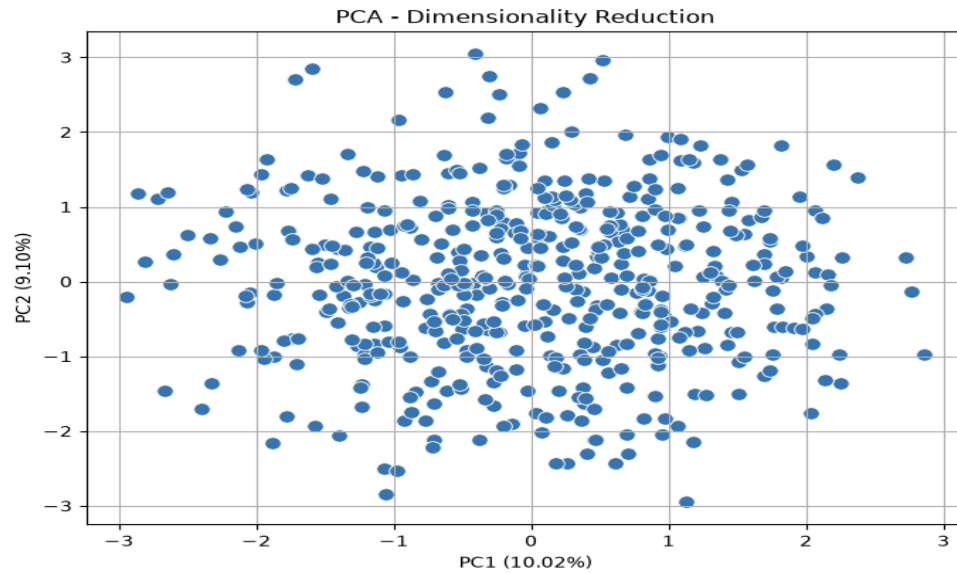
Cluster Counts:
Cluster
2 264
0 144
1 92
Name: count, dtype: int64



Silhouette Score:
0.0965

Explained Variance Ratio:
[0.10023767 0.09101315]

Total Explained Variance: 0.1913



Silhouette Score after PCA: 0.3529
Dataset Size: (500, 15)
Number of Clusters: 3
K-Means Silhouette Score: 0.0965
PCA Explained Variance: 0.1913
PCA K-Means Silhouette Score: 0.3529

Result:

K-Means clustering and PCA were successfully implemented on the Pet Adoption dataset. K-Means grouped the pets into clusters based on feature similarity, while PCA reduced the feature dimensions to two principal components for visualization. The clustering quality was evaluated using the **Silhouette Score**.